Amino acid catabolism: the urea cycle

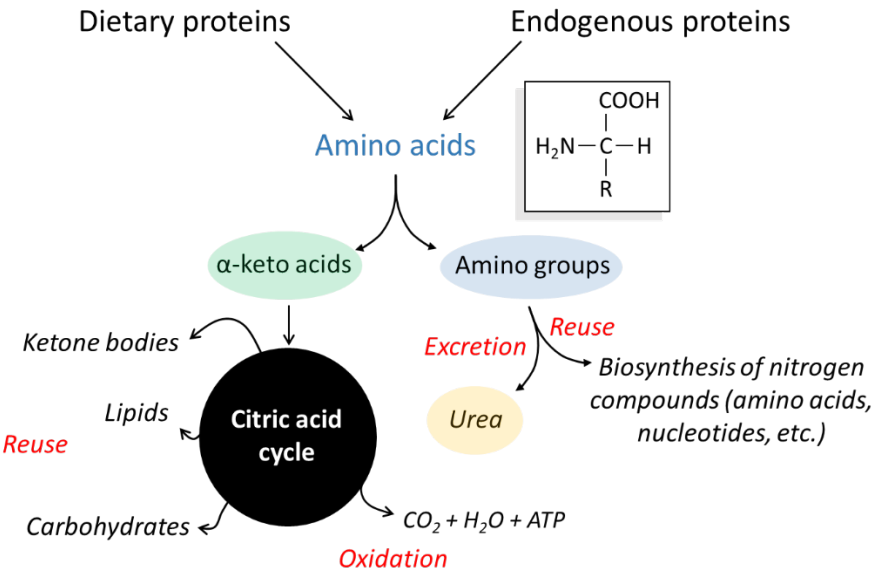
Metabolism of **nitrogen-containing biomolecules**, such as proteins and nucleic acids, differs significantly from that of carbohydrates and lipids. Degradation of proteins and nucleic acids yields nitrogen products that are toxic and have to be eliminated. Our organism needs to be in **nitrogen balance**, where daily intake of nitrogen through the diet equals that lost through excretion. Consequently, unlike carbohydrates and lipids, **amino acids are not stored in a specialized reserve form**. Body proteins constitute a large pool of amino acids, but they have structural and functional roles and are continuously synthesized and degraded. Therefore, excess amino acids cannot simply be stored and must be metabolized.

In animals, degradation of amino acids takes place in the following circumstances:

1. During intracellular **protein turnover**, some amino acids are reused, while others undergo degradation.
2. After a **protein-rich meal**, if ingested amino acids exceed the body's needs, the excess amino acids cannot be stored and are catabolized.
3. In some pathological conditions, e.g. **diabetes mellitus**, or when dietary protein is insufficient, e.g. during **starvation**, proteins (such as muscle proteins) are broken down for the production of energy and precursors.

General strategy for amino acid catabolism

For most amino acids, the first step in their degradation is the **separation of the amino group from the carbon skeleton**. The resulting α-keto acids undergo oxidation to CO_2 and H_2O, or provide intermediates for the synthesis of carbohydrates, fatty acids or ketone bodies. On the other hand, the amino group can be used for the biosynthesis of nitrogenous compounds, or catabolized to urea, which will be excreted in the urine.

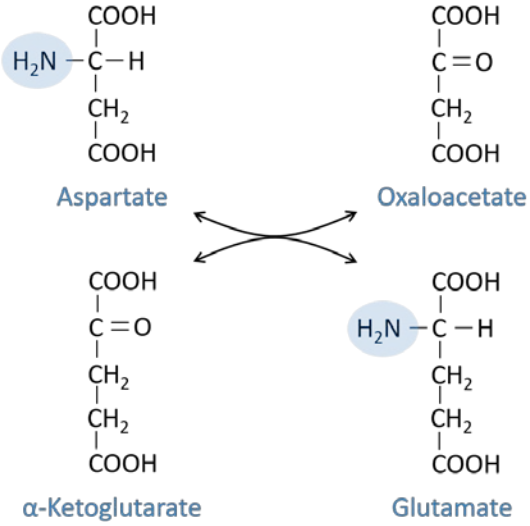


Transamination and transport of amino groups to the liver

For most amino acids, **transamination** is the first step in the removal of the amino group. In this process, nitrogen is transferred as an amino group from the original amino acid to α-ketoglutarate. In that way, the amino acid is converted to its corresponding α-keto acid and α-ketoglutarate to glutamate. **Transaminations are reversible**, so, they are involved in both amino acid degradation and synthesis.



The enzymes that catalyze these reactions are known as ***transaminases*** (or ***aminotransferases***). Different aminotransferases are found in the cytosol and mitochondria. For most of these reactions, α-ketoglutarate and glutamate serve as one of the α-keto acid/amino acid pairs. **Pyridoxal phosphate (PLP)**, which derives from pyridoxine or vitamin B_6, is the required cofactor for these reactions.



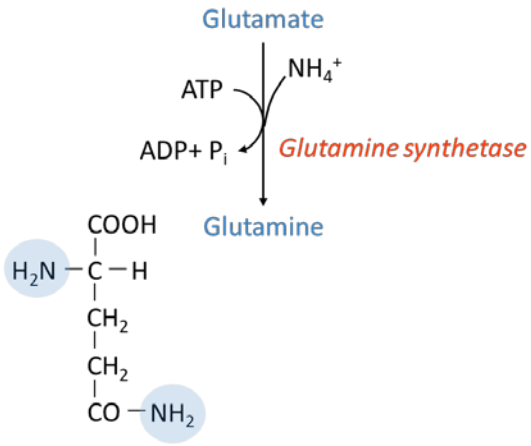
In this example, the amino acid aspartate is transaminated to form its corresponding α-keto acid, oxaloacetate. In the process, the amino group is transferred to α-ketoglutarate, which is converted to its corresponding amino acid, glutamate. The transaminase that catalyzes this reaction is known as **aspartate transaminase *(AST)*** *or glutamate-oxaloacetate transaminase (GOT). AST is one of the most abundant transaminases in our cells.*

Amino acid degradation occurs in all tissues. However, the urea cycle, in which the nitrogen from the amino group is metabolized for its elimination, takes place primarily in the liver. Thus, there are mechanisms to transport the nitrogen to the liver. **Glutamine and alanine are the major carriers of nitrogen in the blood**. Glutamine is the major nitrogen carrier from most extrahepatic tissues, whereas alanine is primarily exported from muscle cells.

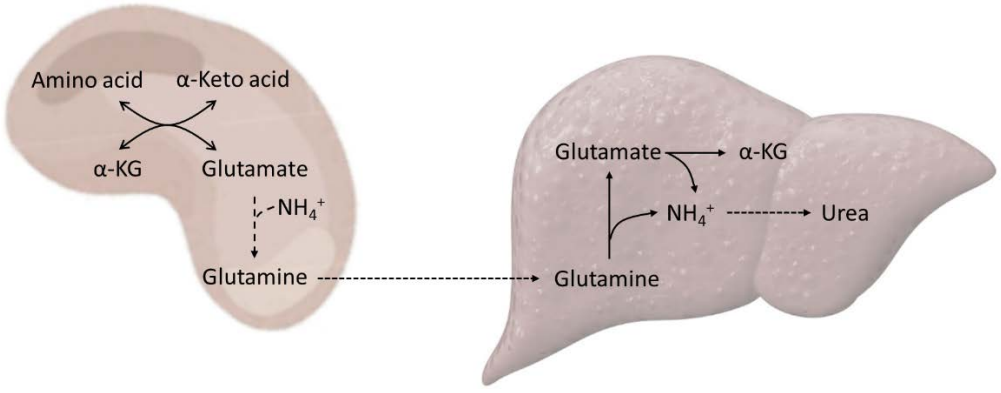
Transport of glutamine to the liver

In extrahepatic tissues, amino groups obtained in the process of amino acid degradation have to be exported into the blood as a non-toxic molecule, and transported to the liver for their disposal. **In most tissues, the transport of amino groups to the liver is carried out by glutamine**. In fact, glutamine is the most abundant amino acid in the blood.

Through transamination reactions, amino groups from many amino acids are transferred to α-ketoglutarate, producing glutamate. In this way, **glutamate acts as a major collector of amino nitrogen**. In extrahepatic tissues, glutamate can incorporate an additional ammonium ion in a reaction catalyzed by *glutamine synthetase*. This ATP-dependent reaction produces glutamine, a nontoxic carrier of nitrogen in the blood.



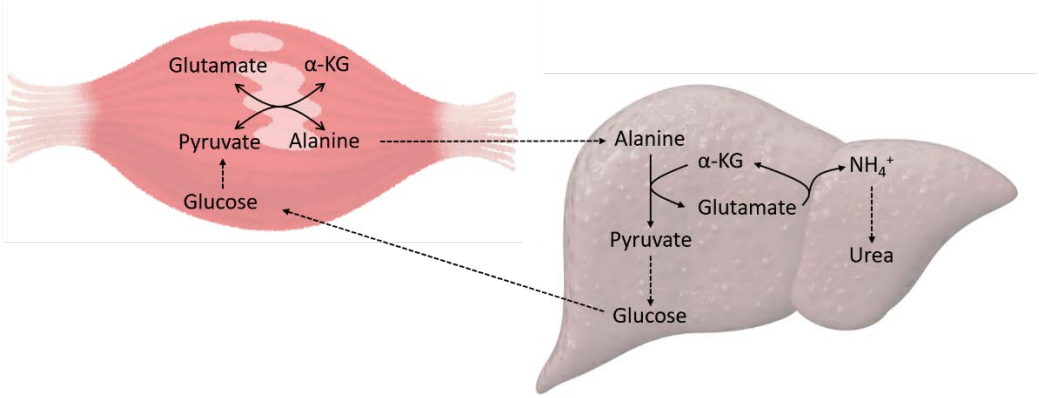
Glutamine travels to the liver for processing. Once in the liver, glutaminase catalyzes the conversion of glutamine to glutamate and NH_4^+ in the mitochondria. The released NH_4^+ can enter the urea cycle. The amino group of glutamate can either be released as NH_4^+ by *glutamate dehydrogenase* or transferred to oxaloacetate to form aspartate. Thus, glutamine can contribute nitrogen to both nitrogen inputs of the urea cycle.



Transport of alanine to the liver

Alanine can also be used for transport of amino groups to the liver in a non-toxic form, mainly by muscle. Once glutamate is formed by transamination, it can transfer its α-amino group to pyruvate by the action of *alanine aminotransferase* (ALT), also known as *glutamate-pyruvate transaminase* (GPT). Because the muscle is metabolizing glucose through glycolysis, pyruvate is readily available in the muscle. Alanine passes into the blood and travels to the liver. Alanine is the second most abundant amino acid in the blood.

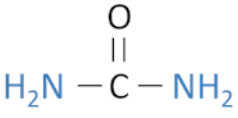
In the cytosol of hepatocytes, ALT catalyzes the transfer of the amino group from alanine to α-ketoglutarate, forming pyruvate and glutamate. Glutamate can then enter mitochondria, where NH_4^+ is released. Pyruvate is either oxidized or converted to glucose via gluconeogenesis. In the latter case, the newly formed glucose can enter the blood for delivery back to the muscle. This cycle of moving carbons and nitrogen between the muscle and the liver is known as the **glucose-alanine cycle**.



The urea cycle

Once ammonia from amino acid degradation is deposited in the mitochondria of hepatocytes, it has to be eliminated to avoid its toxicity. In animals, there are different strategies to excrete nitrogen, which depend on animal physiology and habitat. For most aquatic animals (e.g. fish), which have unlimited quantities of water, ammonia simply dissolves in the water and diffuses away. These animals are called ammonotelic animals. Terrestrial animals must limit the quantities of water losses and therefore have developed biological systems to reduce water loss while eliminating nitrogen. Mammals excrete most amino nitrogen in the form of urea. Urea is highly soluble in water, has low toxicity, and does not significantly affect pH. Mammals are therefore classified as ureotelic animals. Birds and terrestrial reptiles further reduce the water loss and convert ammonia to uric acid (uricotelic animals). Uric acid is insoluble, precipitates, and can be excreted without water loss.

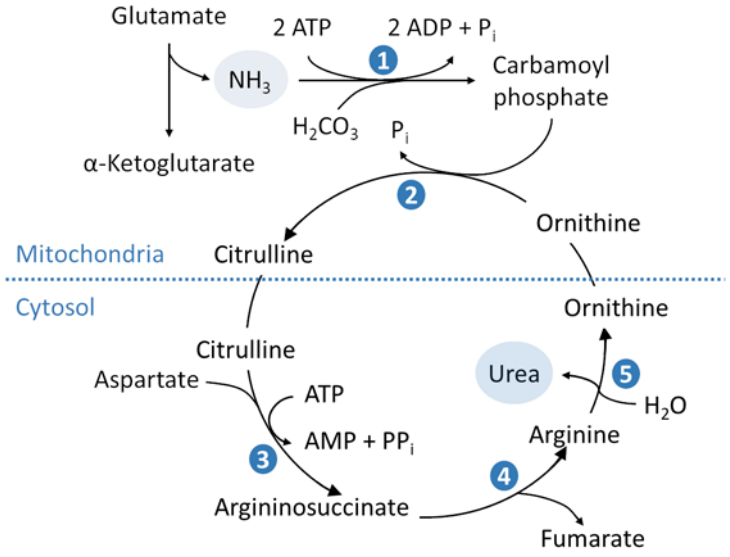
Humans are ureotelic animals; most excess amino nitrogen is eliminated as urea. In each urea molecule, two nitrogen atoms are eliminated.



The synthesis of urea takes place in a pathway known as the **urea cycle**. This cycle was described by Hans Krebs and Kurt Henseleit in 1932 (before Hans Krebs described the citric acid cycle).

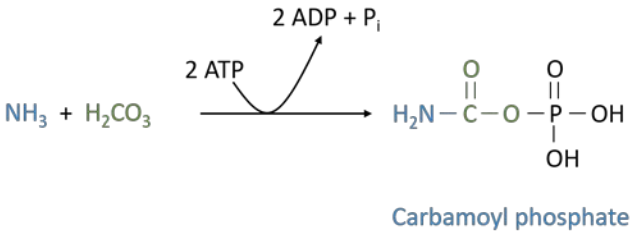
The urea cycle takes place **almost exclusively in the liver**. Once formed, urea passes into the bloodstream and travels to the kidneys, where it is excreted in the urine.

The urea cycle takes place in two cellular compartments, mitochondria and cytosol. The process is composed of five steps, and starts with the ammonia deposited in the mitochondria.

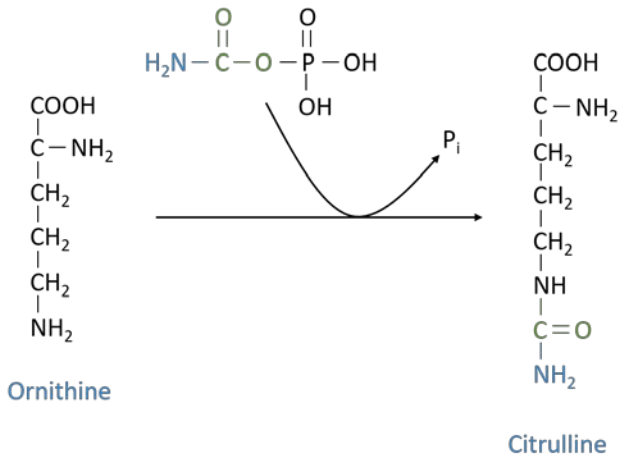


Reactions of the urea cycle

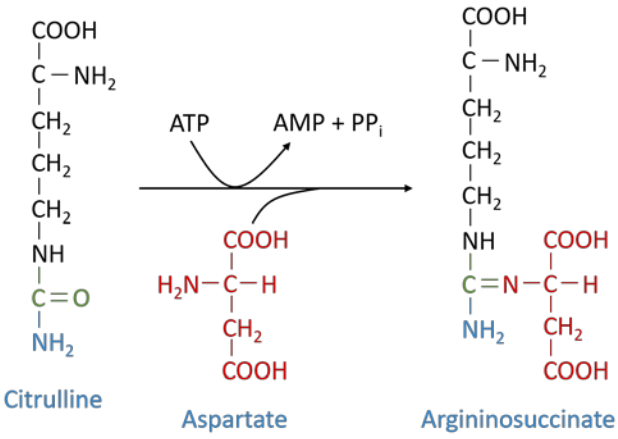
1. Synthesis of carbamoyl phosphate: in the mitochondrial matrix, NH_4^+ and bicarbonate (HCO_3^-) are used to synthesize carbamoyl phosphate. This process requires two ATP molecules and is catalyzed by ***carbamoyl phosphate synthetase I***.



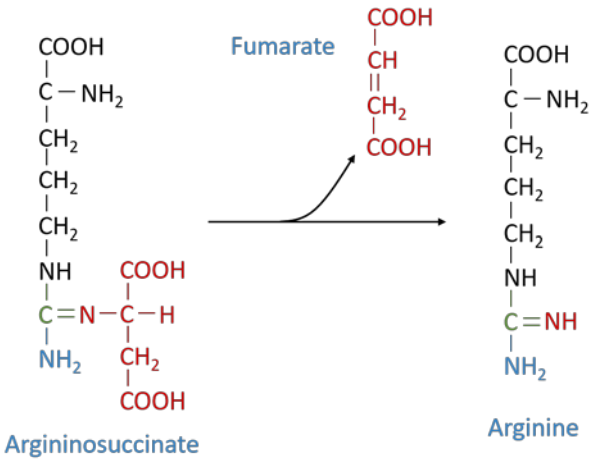
2. Condensation between carbamoyl phosphate and ornithine: carbamoyl phosphate enters the urea cycle by donating its carbamoyl group to the amino acid ornithine to form the amino acid citrulline, with the release of inorganic phosphate. The reaction is catalyzed by ***ornithine transcarbamoylase*** **(OTC)**. Citrulline leaves the mitochondria to the cytosol using a specific carrier.



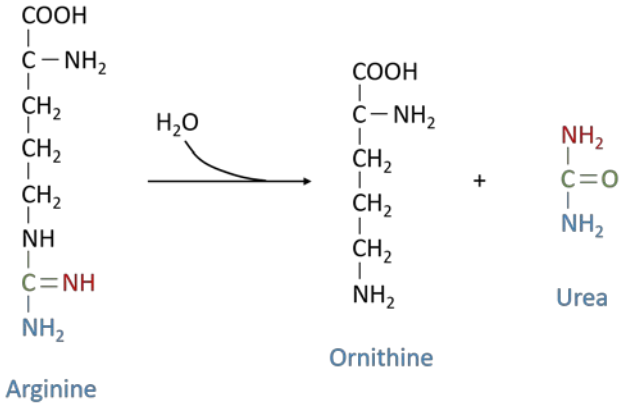
3. Condensation between citrulline and aspartate: the carbonyl group of citrulline reacts with aspartate forming argininosuccinate. This cytosolic reaction is catalyzed by ***argininosuccinate synthetase*** and requires the hydrolysis of two high-energy phosphate bonds. In that way, the second nitrogen atom is introduced into the cycle. Note that aspartate can be generated in the mitochondria by transamination from oxaloacetate in a reaction catalyzed by AST. Therefore, ultimately, **both nitrogen atoms incorporated into urea can originate from glutamate**: one as NH_4^+ and the other through aspartate.



4. Cleavage of argininosuccinate: argininosuccinate is cleaved in a reaction catalyzed by ***argininosuccinate lyase*** to form arginine and fumarate. Fumarate can be converted to malate and then to oxaloacetate. Oxaloacetate can undergo transamination to produce aspartate, which returns nitrogen to the urea cycle. This provides an important metabolic connection between the urea cycle and the citric acid cycle.



5. Cleavage of arginine: the cytosolic enzyme ***arginase I*** catalyzes the hydrolysis of arginine to yield urea and ornithine. Arginase I, the isoform involved in the urea cycle, is expressed predominantly in the liver. Ornithine is transported back into the mitochondria to initiate another round of the urea cycle.



Overall balance of the urea cycle

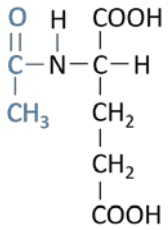
$$CO_2 + NH_4^+ + 3\ ATP + aspartate + 2\ H_2O \rightarrow urea + 2\ ADP + AMP + 4\ P_i + fumarate$$

Urea synthesis is energetically expensive for the organism.

Regulation of the urea cycle

The activity of the urea cycle is regulated at two levels:

1. Long-term regulation: when the diet is rich in protein, the **rate of synthesis** of liver enzymes involved in the urea cycle is increased. This also occurs during prolonged starvation, when muscle protein begins to be used.
2. Short-term regulation: ***carbamoyl phosphate synthetase I*** **is allosterically activated by N-acetylglutamate**.



N-acetylglutamate is synthesized from acetyl-CoA and glutamate in a reaction catalyzed by ***N-acetylglutamate synthase***. Arginine activates this enzyme, increasing N-acetylglutamate production. Thus, increased amino acid catabolism stimulates the urea cycle by increasing the concentration of its obligatory allosteric activator.

Defects in the urea cycle

Hyperammonemia is a pathological condition in which high levels of ammonia are present in the blood. Ammonia is neurotoxic, causing lethargy, tremors, slurred speech, blurred vision, vomiting, coma, and death.

Hyperammonemia can be caused by:

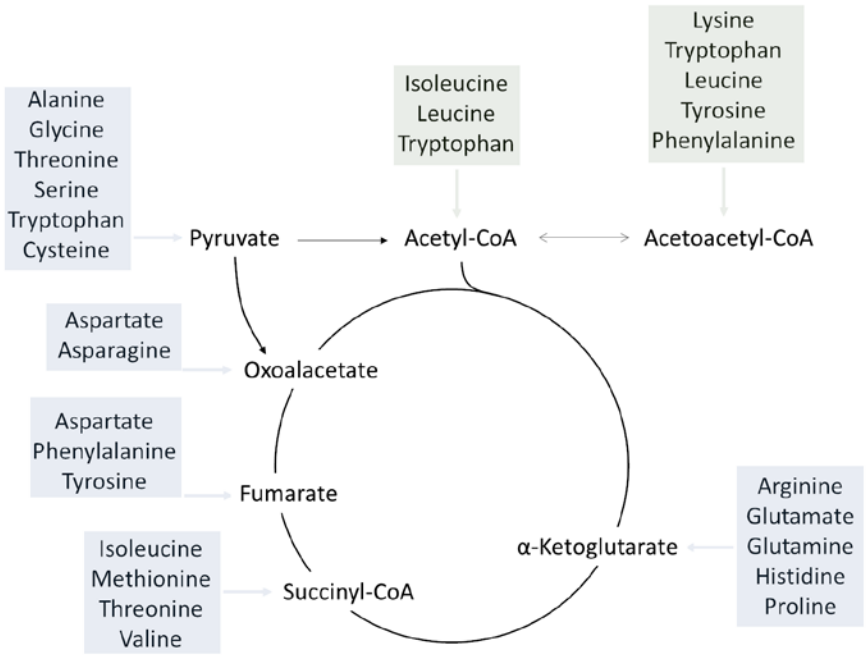
- Hepatic damage: from alcohol, toxic chemicals, hepatitis, etc.
- Genetic defects affecting enzymes or transporters of the urea cycle: the most frequent deficiency affects *ornithine transcarbamoylase* (OTC), but deficiencies can also be found in *carbamoyl phosphate synthetase I*, *ornithine translocase*, *argininosuccinate synthetase*, *argininosuccinate lyase*, *arginase*, and even *N-acetylglutamate synthase*, the enzyme responsible for the synthesis of the allosteric activator of the pathway.

Individuals with urea cycle disorders cannot tolerate a diet rich in protein. They generally require controlled protein intake that provides adequate amounts of essential amino acids while limiting nitrogen load. There are also specific therapies for particular enzyme deficiencies.

Catabolism of α-keto acids

The carbon skeletons derived from the transamination of amino acids are degraded or can be diverted to the synthesis of carbohydrates or fatty acids. In this respect, amino acids fall into two categories: glucogenic and ketogenic.

Glucogenic amino acids are those that give rise to pyruvate or citric acid cycle intermediates, which are precursors of glucose. **All amino acids, except lysine and leucine, are glucogenic**, at least partially. **Ketogenic amino acids** give rise to acetyl-CoA or acetoacetyl-CoA, neither of which can be used for glucose production. Lysine and leucine are the only amino acids that are purely ketogenic.



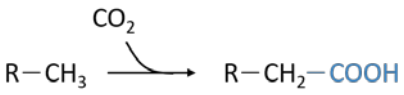
Cofactors involved in amino acid catabolism

Several enzyme cofactors participate in the degradation of the carbon skeleton of amino acids. Many of them are also involved in other metabolic pathways, especially those related to amino acid synthesis and nucleotide metabolism.

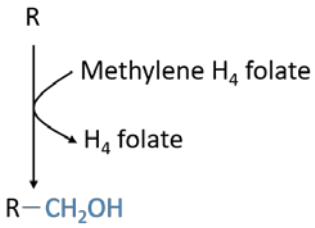
Some of the most relevant cofactors are:

1. **Pyridoxal phosphate (PLP)**: derived from vitamin B_6, is used in **transamination** reactions, but also in isomerizations, modifications of amino acid side chains, decarboxylations, etc.

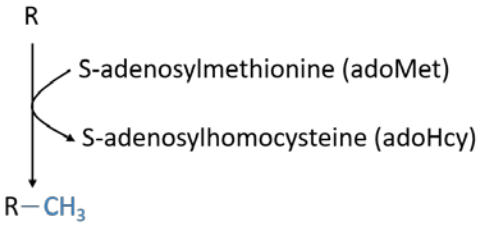
2. **Biotin**: vitamin B_7, is a cofactor that participates in **carboxylation** reactions. Besides reactions in amino acid metabolism, biotin is also a cofactor for *acetyl-CoA carboxylase*, *propionyl-CoA carboxylase*, *pyruvate carboxylase*, etc.



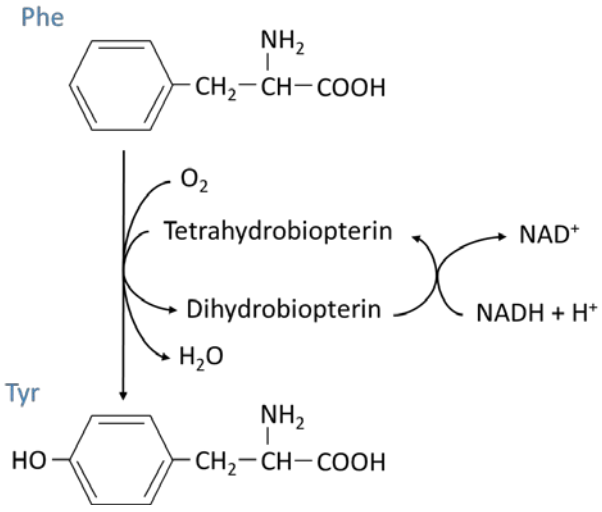
3. **Tetrahydrofolates (THF)**: derived from folate (vitamin B_9), participate in the **transfer of one-carbon units in different oxidation states**, including methyl, methylene, methenyl, and formyl groups.



4. **S-adenosylmethionine (SAM or AdoMet)**: is the major biological **methyl-group donor**. Apart from its role in amino acid metabolism, SAM serves as the methyl-group donor for many methyltransferases, including those involved in DNA methylation.



5. **Tetrahydrobiopterin (BH_4)**: participates in **oxidation-reduction reactions**.



Phenylalanine hydroxylase *catalyzes the conversion of phenylalanine to tyrosine and requires tetrahydrobiopterin (BH_4) as a cofactor. During the reaction, one oxygen atom from O_2 is incorporated into phenylalanine, while the other is reduced to H_2O.* Phenylalanine hydroxylase *belongs to the class of enzymes known as mixed-function oxidases, which catalyze simultaneously the hydroxylation of a substrate by an oxygen atom of O_2, and the reduction of the other oxygen atom to H_2O. BH_4 provides the reducing equivalents required for the reaction and is subsequently regenerated. BH_4 is regenerated from its oxidized form through reactions that ultimately use NADH as a reducing agent.*

Genetic defects in amino acid catabolism

Several genetic disorders affect the catabolism of amino acids. Some examples are phenylketonuria, tyrosinemias, homocystinuria, non-ketotic hyperglycinemia or maple syrup urine disease. They can be diagnosed by analyzing amino acid concentrations in body fluids.

One of the most common inherited disorders of amino acid metabolism is **phenylketonuria** (PKU). Classic PKU is caused by a **genetic deficiency of *phenylalanine hydroxylase***, resulting in the accumulation of phenylalanine to neurotoxic concentrations. If untreated, the disease can cause severe neurological damage, seizures, and intellectual disability. Early diagnosis through newborn screening (heel prick test) and dietary restriction of phenylalanine can prevent most of these complications.

Author: Ruben Pio

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